

Geometry
Unit 7
Lesson H2 Practice

Name Answer Key
Date _____
Period _____

In $\triangle GHS$, $m\angle HGS = 22^\circ$, $HG = 27$, and $SG = 14.9$. Determine the measurement of each part of $\triangle GHS$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
1.	$\angle SHG$	23°
2.	$\angle HSG$	135°
3.	$\overline{HS}$	14.3

$$1. \quad 14.9^2 = 27^2 + 14.32^2 - 2(27)(14.32) \cos B$$

$$773.28 \cos B = 712.0524$$

$$B = 22.954^\circ$$

$$2. \quad 27^2 = 14.9^2 + 14.32^2 - 2(14.9)(14.32) \cos a$$

$$426.736 \cos a = -301.928$$

$$a = 135.03$$

$$3. \quad x^2 = 27^2 + 14.9^2 - 2(27)(14.9) \cos 22 = 729 + 222.01 - 746.012 = 204.998$$

$$x = 14.32$$

In $\triangle PXN$, $PX = 17$, $PN = 28$, and $XN = 30$. Determine the measurement of each part of $\triangle PXN$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
4.	$\angle PXN$	66.6°
5.	$\angle PNX$	33.9°
6.	$\angle XPN$	79.5°

$$4. \quad 28^2 = 30^2 + 17^2 - 2(30)(17) \cos x$$

$$1020 \cos x = 405$$

$$x = 66.6$$

$$5. \quad 17^2 = 30^2 + 28^2 - 2(30)(28) \cos n$$

$$1680 \cos n = 1395$$

$$n = 33.9$$

$$6. \quad 30^2 = 17^2 + 28^2 - 2(17)(28) \cos p$$

$$952 \cos p = 173$$

$$p = 79.5$$

Austyn is checking a water trough in her pasture before walking to the barn. She enters the pasture through a gate near her house. The distance from the gate, point G , to the water trough, point W , is 52 yards. The distance from the gate to the barn, point N , is 37.5 yards. The distance from the water trough to the barn is 31 yards. Determine each angle measure.

7.
8.
9.

Angle	Measurement
$\angle NGW$	36°
$\angle GWN$	46°
$\angle WNG$	98°

10. In $\triangle DTF$, $DF = 10$, $TD = 42$, and $TF = 46$. Complete the statement about $\triangle DTF$.

$\triangle DTF$ is a

☐ right
☐ acute
☒ obtuse
☐ right

triangle because

☒ one of the angles is
☐ all of the angles are

☐ equal to 90° .
☐ less than 90° .
☒ greater than 90° .

$\overline{FT}$ is longest segment so $\angle FDT$ is largest angle.

$$46^2 = 10^2 + 42^2 - 2(10)(42) \cos b$$

$$\cos b = \frac{-252}{840}$$

$$b = 107.46^\circ$$